

Better Batteries for Your EV



How Steady Speeds Lead to Better Batteries for Your EV



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Published Apr 1, 2024

Ever wondered why car manufacturers quote a specific range for their electric vehicles (EVs)? Achieving that target relies heavily on a rather unexciting factor: constant speed. While cruising down the highway at a steady pace might seem mundane, it plays a crucial role in both how EVs are rated and how well their batteries perform in the long run.

Building Trust Through Predictable Testing

Imagine trying to measure gasoline mileage for a car if every test involved a different route with varying traffic and speeds. It would be nearly impossible to get consistent results. This is why EV manufacturers rely on standardized testing procedures that involve maintaining a constant speed. This controlled environment allows for reliable and repeatable testing, enabling them to accurately assess battery performance and confidently advertise a **commitment range**.

This predictability is crucial for building consumer trust. By showcasing a dependable range under controlled conditions, manufacturers assure potential buyers of their vehicle's capabilities. You, as a consumer, can rely on the advertised range as a benchmark for how far you can expect to travel on a single charge under ideal circumstances.

Optimizing for the Steady State

Think of an EV's powertrain as a complex orchestra. The electric motor, battery, and control systems all need to work together in perfect harmony to achieve maximum efficiency. When engineers know the typical driving scenario (constant speed in this case), they can fine-tune this orchestra to perform its best at that particular speed. This translates to a battery that excels at delivering power consistently, maximizing the advertised range.

Just like how a singer struggles to hit high notes while running, an EV battery experiences stress when constantly switching between acceleration and deceleration. By optimizing for a steady pace, engineers can create a battery system that thrives under those conditions, delivering optimal performance and efficiency.

Fine-Tuning the Battery's Maestro: The BMS

The Battery Management System (BMS) acts as the conductor of the battery orchestra. It constantly monitors and regulates factors like cell voltage and temperature to ensure optimal performance and longevity. With a focus on constant-speed efficiency, engineers can program the BMS to optimize battery usage and charging specifically for that driving scenario. This allows the BMS to make smarter decisions about power delivery and heat management, ultimately contributing to a healthier battery.

The Real World: A Balance Between Speed and Efficiency

Of course, real-world driving rarely involves constant cruising on a highway. Traffic lights, stop signs, and varying speeds are all part of the daily commute. While maintaining a steady speed isn't always possible, the insights gained from constant-speed testing provide a valuable benchmark. This benchmark allows OEMs to design EVs that are efficient at a speed achievable in many driving situations.

It's important to remember that the actual range you experience will vary depending on factors like terrain, temperature, and your driving style. However, understanding the importance of constant speed in EV design and testing sheds light on how it ultimately translates to a better battery experience for you.

The Future is Looking Bright

Battery technology is constantly evolving, with manufacturers striving to improve battery chemistry and design to reduce attenuation (gradual loss of capacity). Additionally, modern EVs come equipped with sophisticated BMS systems that can regulate charging currents and temperatures to minimize attenuation even during fast charging.

While slow charging offers advantages for battery health, the ideal charging strategy depends on your individual needs. If you prioritize long-term battery health and have ample charging time, slow charging is a great option. However, a mix of slow and fast charging can be a practical approach if time is a constraint.

So, the next time you're cruising down the road at a steady pace, remember that you're not just enjoying a smooth ride; you're also contributing to a future of efficient and long-lasting EV batteries.